

MTU - Team “Delta L2”

AI-Based "Wingman" for Detecting PPE Violation and
Emergency Situations on the Factory Floor

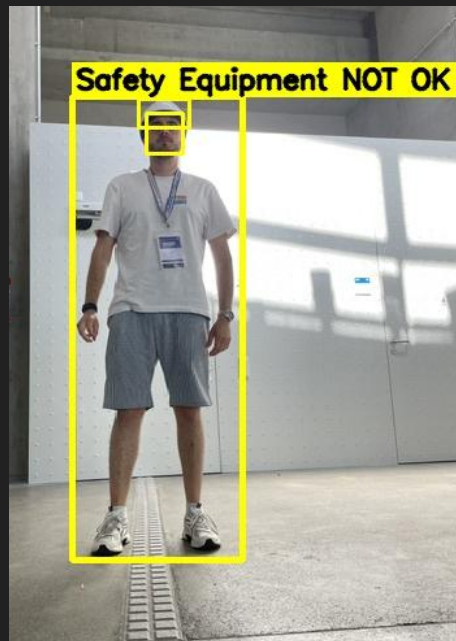
Problem

- Workers on factory floors may be required to wear personal protective equipment (PPE), such as hard hats and safety vests.
- Workers who fall or are incapacitated may go unnoticed for the first critical minutes of the incident

Our Solution

- AI based person detection and classification: Safety Equipment OK or NOK, Emergency
- Providing a dashboard as an intuitive overview
- Anonymization for compliance with the General Data Protection Regulation

Example



PPE Violation

Not wearing hardhat or not wearing a safety vest?

Emergency Situation
Is a Person lying on the Ground?



Architecture Overview



 FastAPI

 SQLite

Backend

Ingest events, manage Images and
prepare data for presentation

2

Frontend

3

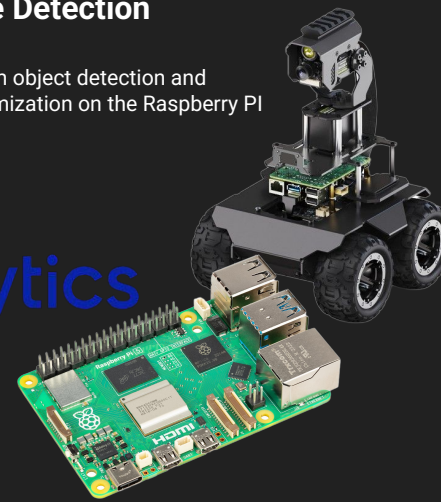
Real time monitoring of events,
analytics for operator

Edge Detection

1

Custom object detection and
anonymization on the Raspberry PI

 ultralytics
YOLO26



Edge Detection

- Object Detection using custom YOLO26 models
- Optimized for problem specific classes
- More robust anonymization

Combined PPE Datasets

Detection of hardhats and safety vests



Face Detection Dataset

Model for face detection, more accurate for anonymization



Wingman Model

Detects Person, Heads, Hard Hats, Vests with High accuracy

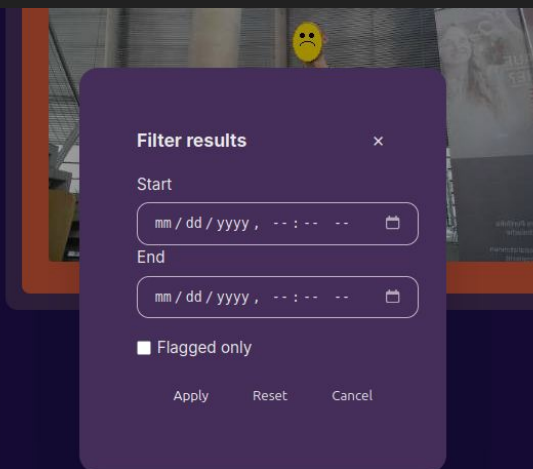
Backend

- Central processing of events based on FastAPI
- Stores events and anonymized images
- Prepares data for presentation
- REST-API



Dashboard

- Live Events
- Statistics
- Filtering



WINGMAN dashboard

116 Events

until now



Filter

Missing Hardhat

10:48 21.6.2026

No Safety Vest

10:48 21.6.2026

Missing Hardhat

10:48 21.6.2026

Detection of the last 7 days

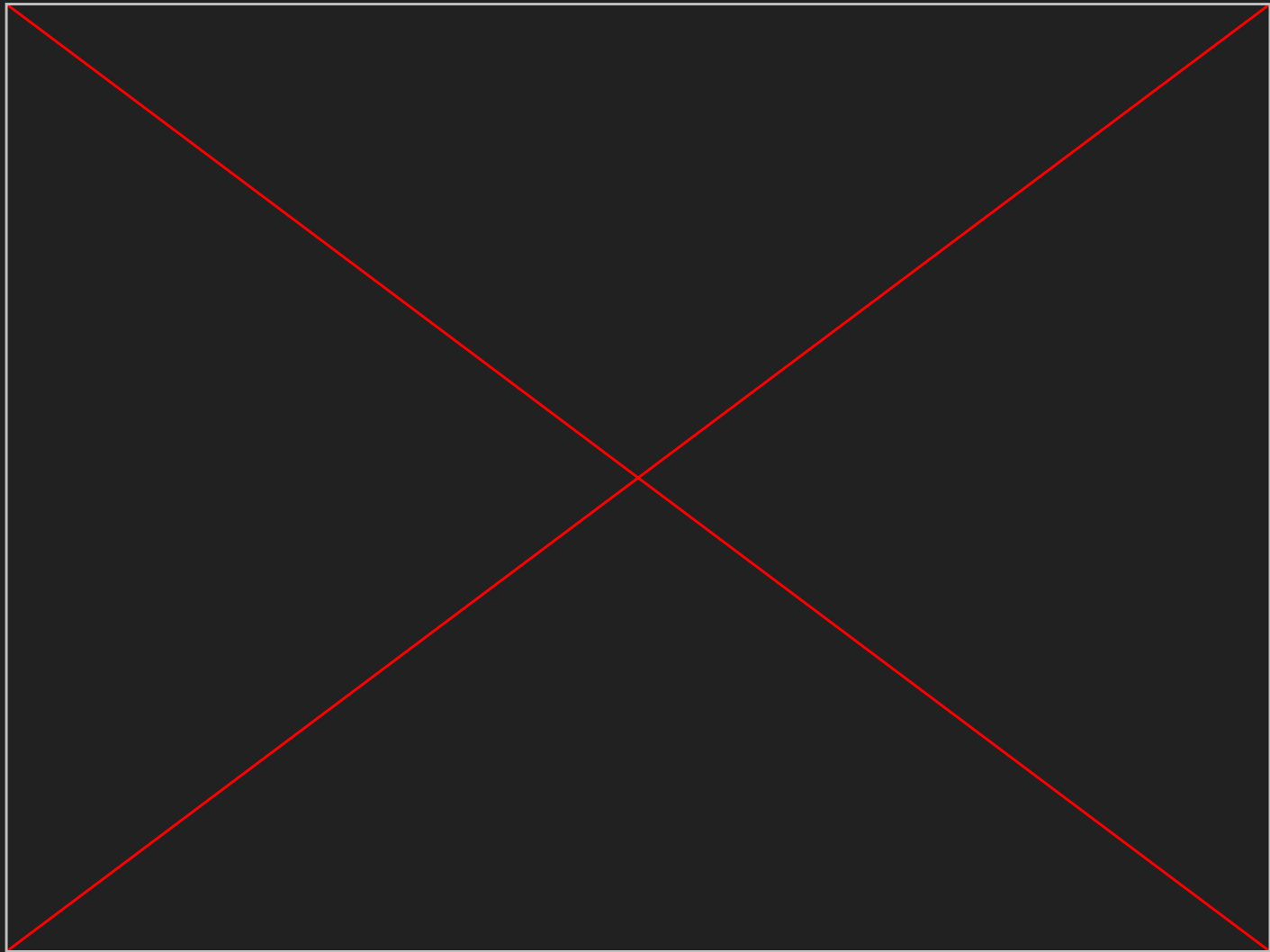


■ Missing Hardhat: 32
■ No Safety Vest: 8
■ Emergency: 4

Event Details



Demo



Summary

Self set baseline was achieved:

1. AI model on edge device
2. Backend receiving events and images
3. Frontend with the dashboard

... and there are a lot ideas for further development and improvement

→ Ask us about it 🤔

Learnings - TOP 3

1. Priority Management → Only focus on the function and the pipeline
2. No overengineering → less custom code, more proven solutions
3. Steep learning curve with the necessity to get up to speed quickly

Thanks to Stefan and Martin for the supervision 🙌

The Repository

Visit our repository on GitHub:

[mtuhack](https://github.com/jacobriege/mtuhack) - <https://github.com/jacobriege/mtuhack>

or

[HackAero/2026_mtu_team_1](https://github.com/HackAero/2026_mtu_team_1) -
https://github.com/HackAero/2026_mtu_team_1

 README

Science Hackathon 2026 - MTU Team 1

AI-Based "Wingman" for Detecting PPE Violation and Emergency Situations

Problem

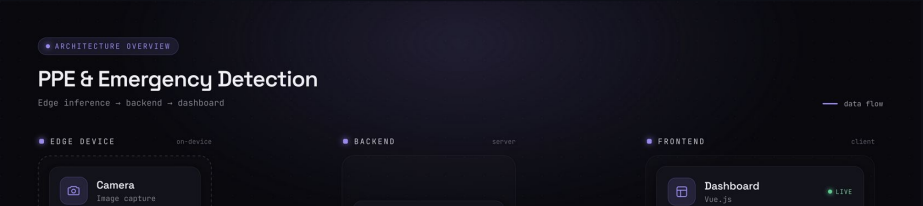
Factory floors require workers to wear personal protective equipment (PPE) — hardhats and safety vests — at all times. Also Workers who fall or are incapacitated may go unnoticed for critical minutes.

Solution

An automated, real-time safety monitoring system that uses computer vision on a rover to patrol a factory floor and detect PPE violations and workers in emergency situations. Violations are anonymized and sent to a central backend, where supervisors can review them through a dashboard.

Implementation

Architecture Overview



The diagram illustrates the system architecture for PPE & Emergency Detection. It is divided into three main components: **EDGE DEVICE** (on-device), **BACKEND** (server), and **FRONTEND** (client). The **EDGE DEVICE** includes a **Camera** for image capture. The **BACKEND** is represented by a box labeled 'server'. The **FRONTEND** includes a **Dashboard** (Vue.js) which is shown as **LIVE**. A **data flow** arrow connects the Edge Device to the Backend, and the Backend to the Frontend. A navigation bar at the top of the diagram shows 'ARCHITECTURE OVERVIEW' as the selected tab. Below the title 'PPE & Emergency Detection', the flow is summarized as 'Edge inference -> backend -> dashboard'.